

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Portfolio Risk-Factor Analysis Using PCA

Dimensionality Reduction for Portfolio Risk Management

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UNDER THE GUIDANCE OF

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Why Portfolio Risk Is Hard to Measure

1

Too Many Assets

Modern portfolios often hold 100+ stocks, each with its own risk pattern.

2

Hidden Overlaps

Sector and market shocks move many holdings together at the same time.

3

Old Tools Don't Scale

Full covariance matrices get noisy, slow, and hard to read as holdings grow.



What We Set Out to Do

1

Simplify Risk

Reduce hundreds of risk signals down to a few key factors

2

Find Patterns

Use PCA to uncover hidden structure inside stock returns

3

Name the Factors

Give each mathematical pattern real financial meaning

4

Visualize It

Prototype a simple dashboard to track portfolio risk

OUR APPROACH

A Five-Step Risk Pipeline



Collect

Gather historical stock returns



Clean

Standardize & remove outliers



Decompose

Apply PCA to find components



Interpret

Map components to real factors



Visualize

Build a risk dashboard

PCA IN PLAIN ENGLISH

Find the few directions where the portfolio moves the most — and keep just those.

Dataset at a Glance

120

Stocks Tracked

10

Sectors Covered

5 Yrs

Daily Price History

Cleaned & Standardized

Missing data filled, outliers capped, and returns standardized before modeling.

Early Insight

Banking and Energy sectors showed the highest weekly volatility.

Stocks Rarely Move Alone

	IT	BANK	ENGY	FMCG
IT	1.00	0.28	0.22	0.12
BANK	0.28	1.00	0.31	0.14
ENGY	0.22	0.31	1.00	0.17
FMCG	0.12	0.14	0.17	1.00

Illustrative correlation levels by sector

Same-sector stocks move together

Correlation above 0.80 within IT and Banking

Different sectors barely interact

Cross-sector correlation typically under 0.35

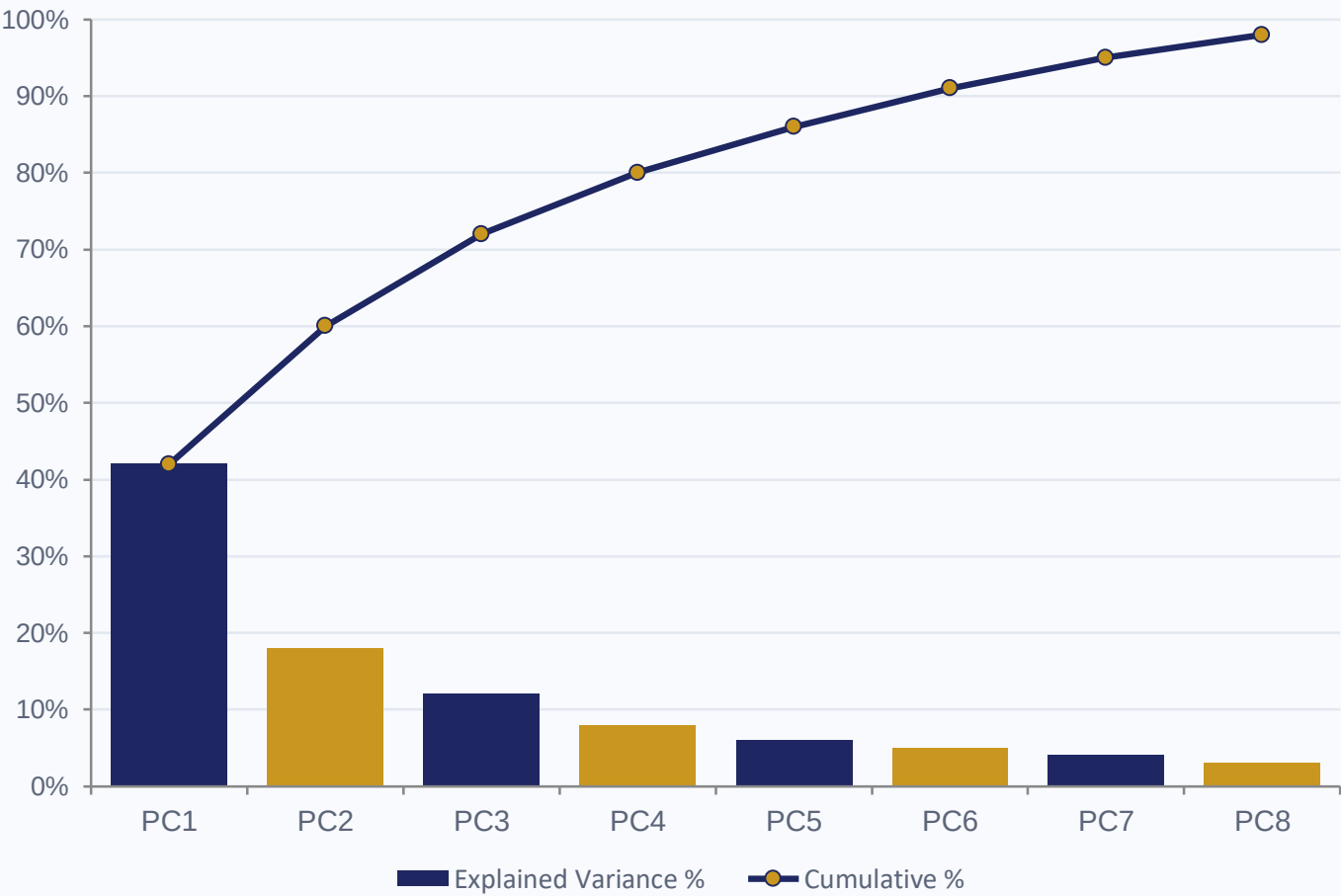
FMCG stands apart

Moves almost independently of other sectors

This overlap is exactly what PCA compresses into fewer factors.

THE RESULT

From 120 Numbers to Just 5



42%

explained by the first factor alone

86%

explained by the top 5 factors

120 → 5

dimensions reduced, risk preserved

Giving the Numbers Financial Meaning



Market

~42%

Broad moves affecting nearly every stock



Sector Rotation

~18%

Financials & energy vs. IT & pharma



Rate Sensitivity

~12%

Driven mainly by banks & real estate



Momentum / Style

~8%

Growth stocks vs. steady, low-volatility names

Prototype: Risk Dashboard Concept



Concept prototype — not a live, connected tool

CONCLUSION

Key Takeaways

120 → 5

dimensions reduced

86%

risk preserved

- Five data-driven factors replace 120 separate risk signals
- Market, Sector Rotation, Rate Sensitivity, and Momentum all carry real meaning
- Faster and easier to interpret than traditional covariance-based tools
- Assumes linear relationships — works best with regular updates

“Simpler models. Clearer risk stories. Faster decisions.”

Thank You